

Track-to-Track Fusion in Linear and Nonlinear Systems^{*}

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Abstract. This chapter starts with a review of the architectures for track-to-track fusion (T2TF). Based on whether the fusion algorithm uses the track estimates from the previous fusion and the configuration of information feedback, T2TF is categorized into six configurations, namely, T2TF with no memory with no, partial and full information feedback, and T2TF with memory with no, partial and full information feedback. The exact algorithms of the above T2TF configurations and the impact of information feedback on fusion accuracy are presented. Although (under the Linear Gaussian assumption) the exact T2TF algorithms yield theoretically consistent fusion results, their major drawback is the need of the crosscovariances of the tracks to be fused, which drastically complicates their implementation. The information matrix fusion (IMF) is a special case of T2TF with memory. Although it is heuristic when not conducted at full rate, it was shown to have consistent and near optimal fusion performance for practical tracking scenarios. Due to its simplicity, it is a good candidate for practical tracking systems. For the problem of asynchronous T2TF (AT2TF), a generalized version of the IMF is presented. It supports information feedback for AT2TF in the presence of communication delay, and was shown to have good consistency and close to optimal fusion accuracy. Finally the fusion of heterogenous tracks where the states at the local trackers are nonlinearly related and of different dimension is discussed. For the problem of the fusion of the track from an Interacting Multiple Model (IMM) estimator from an active sensor with the track from a passive sensor, a counterintuitive phenomenon that heterogenous T2TF may have better performance than the centralized measurement-to-track fusion approach (which is the known optimum in the linear case) is demonstrated and explained.

1 Introduction

In tracking applications, when more than one sensor is used to obtain measurements, there are several possible configurations for information processing, which are summarized in Fig. 1. They differ in the sequence in which the data association and tracking are carried out and the information available to the various processors [6].

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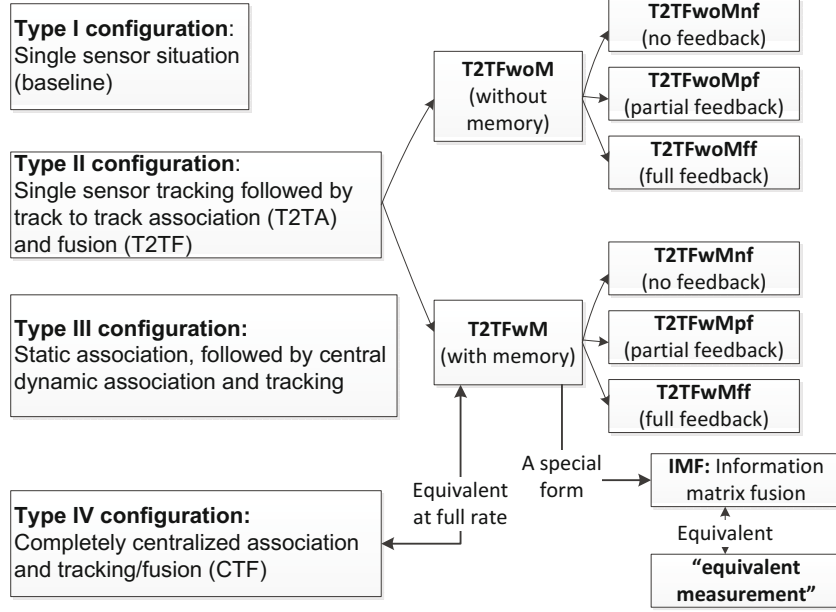


Fig. 1. Configurations for Multisensor Information Processing

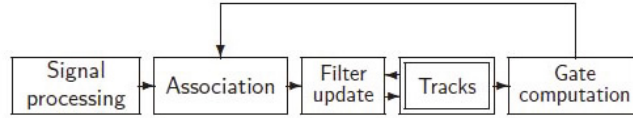


Fig. 2. Type I configuration — single sensor tracking

Type I configuration refers to the (standard) tracking system using a single sensor, which has the flowchart depicted in Fig. 2. In a multisensor situation this corresponds to reporting responsibility (RR). Each sensor operates alone and has responsibility for a certain sector of the surveillance region – no fusion of the data (measurements or tracks) from the multiple sensors is done. As targets move from one sector to another, they are handed over – handoff – in a manner that depends on the system. Generally, the mechanism is to assign responsibility to the sensor with the highest expected accuracy, although workload and communication constraints can also play a role.

Type III configuration is the static intersensor association and fusion followed by central processing which consists of two stages. In the first stage, the measurements from the various sensors, assumed to be from the same time, i.e., the sensors are assumed synchronized, are first associated and fused. This is a static intersensor measurement association and centralized measurement fusion (CMF)

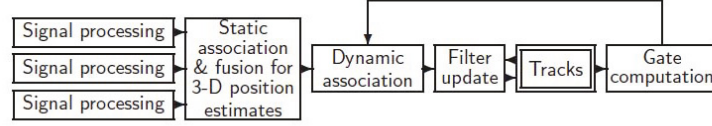


Fig. 3. Type III configuration — static association, extraction of composite measurements (fusion), followed by dynamic association/tracking

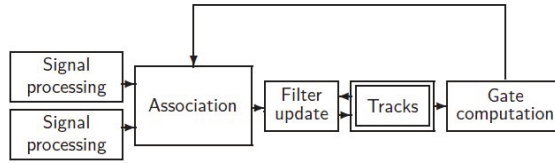


Fig. 4. Type IV configuration — centralized multisensor tracking (Centralized Tracker/Fuser — CTF)

that yields composite measurements, also known as supermeasurements. Then in the second stage, these composite measurements are processed by a (central) dynamic association and tracking algorithm. Fig. 3 shows such a configuration for 3 sensors.

The Type IV configuration, which is the centralized tracking (also called centralized tracker/fuser — CTF), is depicted in Fig. 4. In this configuration all the measurements are sent to the center, which carries out the association with all the available information and then uses these measurements to update the tracks. Since this configuration uses the maximum available information, it will provide (subject to the limitations of the specific data association algorithm it uses) the best results [6]. In the absence of the need for data association, it will yield, for linear systems, the globally optimal estimates [6]. As it will be shown later, in a nonlinear problem with heterogenous trackers, HT2TF (heterogenous T2TF) can be superior to CTF.

The Type II configuration, which is the main subject of this chapter, is the single sensor tracking followed by track fusion. Its importance stems from the fact that it can run at a low rate, e.g., on demand. This is important in situations where the communication bandwidth from the sensors to the fusion center is limited. This configuration is distributed and decentralized, where each sensor has its own information processor — local data associator/tracker — and yields full tracks. A Fusion Center (FC) carries out the association and fusion of the local tracks into system tracks. These steps are designated as Track-to-Track Association (T2TA) and Track to Track Fusion (T2TF), respectively. The Type II configuration is very important in distributed tracking systems. Compared to

the CTF, which requires the transmission of local sensor measurements to the FC at the full rate, T2TF can be conducted at much lower rates, which can significantly reduce communication requirements.

As shown in Fig. 1, depending on whether the fusion algorithm uses the track estimates from the previous fusion and the configuration of information feedback, T2TF configurations are further categorized as: T2TF with no memory with no information feedback (T2TFwoMnf), partial information feedback (T2TFwoMpf) and full information feedback (T2TFwoMff), and T2TF with memory with no information feedback (T2TFwMnf), partial information feedback (T2TFwMpf) and full information feedback (T2TFwMff). These configurations as well as the impact of information feedback on the fusion accuracy are discussed in detail in Sect. 2.

The impact of the crosscorrelation between local estimation errors on fused estimate for T2TFwoMnf is discussed in Sect. 3.

Also shown in Fig. 1, the information matrix fusion (IMF) [10,8,15,11,6] is a special form of T2TFwM. Operating at full rate the IMF is equivalent to the Type IV Configuration, i.e., the CTF, while, at a reduced rate, the IMF is heuristic. However, for the practical range of system process noises levels, it was shown to yield consistent¹ fusion results and close to the optimal fusion accuracy [9]. Also note that the IMF is algebraically equivalent to the equivalent measurement approach [6]. Compared to the exact T2TF fusion algorithms, the IMF has the advantage of not requiring the crosscovariances of the errors of the tracks to be fused, which significantly simplifies the implementation. Sect. 4 reviews the IMF at full and reduced rate, as well as a generalized IMF (GIMF) for the fusion of asynchronous tracks which supports information feedback in the presence of communication delay and was shown to have consistent and close to optimal fusion results [18].

Another special type of T2TF that may occur in practical tracking systems is the fusion of tracks from heterogeneous trackers that are using different state vectors, discussed in Sect. 5. In [19] the HT2TF problem was investigated, where the track from an interacting multiple model (IMM) filter with states in Cartesian coordinates and using an active sensor was fused with the track from a passive sensor with angular states. Counterintuitively, it was shown that when the IMM tracker is involved, HT2TF yielded better performance than the CTF approach. These results are presented in Sect. 6. Sect. 7 summarizes the chapter with concluding remarks.

2 Track-to-Track Fusion Configurations and the Impact of Information Feedback

Depending on whether the track estimates from the previous fusion are used for the current fusion and the configuration of information feedback, T2TF can be categorized as the following configurations:

¹ Its errors were commensurate with its calculated covariance [2].

- T2TFwoM with no information feedback (T2TFwoMnf)
- T2TFwoM with partial information feedback (T2TFwoMpf)
- T2TFwoM with full information feedback (T2TFwoMff)
- T2TFwM with no information feedback (T2TFwMnf)
- T2TFwM with partial information feedback (T2TFwMpf)
- T2TFwM with full information feedback (T2TFwMff)

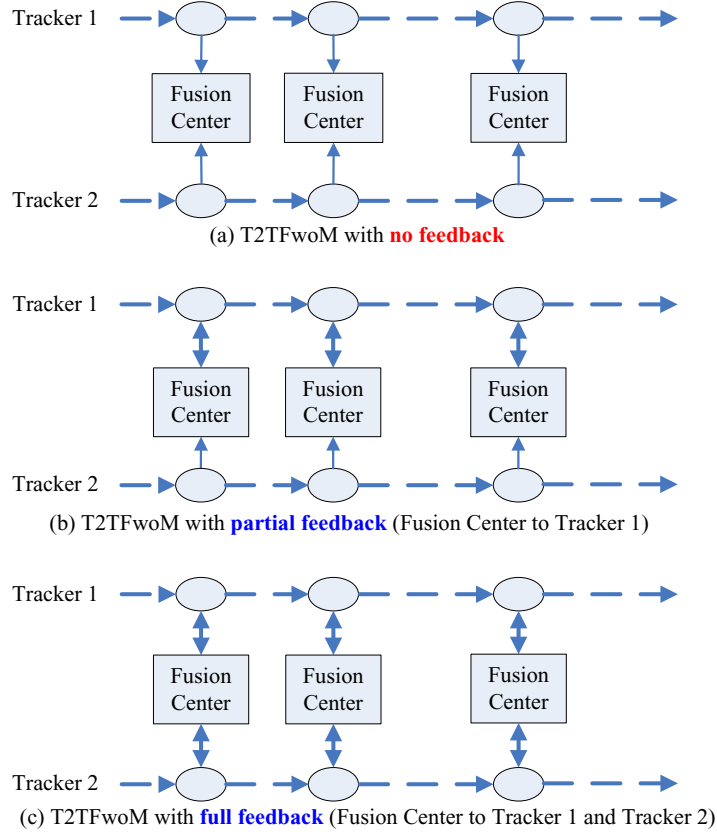


Fig. 5. Information configurations for T2TFwoM

2.1 T2TF without Memory

In T2TFwoM, the FC uses only the current track estimates with no memory of the track estimates from the previous fusion. Fig. 5 illustrates the three information configurations of T2TFwoM, where two local tracks (that pertain to the same target) are fused at certain times. The first configuration is the T2TFwoMnf [5], designated as Config. IIa for multisensor tracking in [1]. As

indicated in Fig. 5(a), the two tracks evolve independently without any information from each other, thus the improved accuracies are achieved only at the fusion times at the FC. The second configuration is the T2TFwoM with partial information feedback (T2TFwoMpf) which belongs to the Config. IIB in [1]. In this case, as shown in Fig. 5(b), track 1 is fused with track 2 and continues with the fused track (feedback) from the FC. However, track 2 does not receive the fused track in view of the partial information feedback. The third configuration is the T2TFwoM with full information feedback (T2TFwoMff), which also belongs to the Type IIB configuration in [1]. As shown in Fig. 5(c), in this case both trackers receive and continue with the fused track.

The exact algorithms for T2TFwoM can be found in [6,16]. The key is to evaluate the crosscovariances between the local tracks. As shown in [3], although the measurements at different local trackers have independent noises, the local tracks are correlated due to common process noises of the target's motion. Ignoring the crosscovariances in T2TF will lead to over-optimistic fused covariance and track inconsistency [6]. In [16] it was shown that, compared to CTF, T2TFwoM always has a certain loss in fusion accuracy and, counterintuitively, information feedback has a negative impact on the accuracy of T2TFwoM.

To illustrate this phenomenon, consider the following generic T2TF example. The target state is defined as $[x \ \dot{x}]'$. The target motion is modeled as the discrete white noise acceleration (DWNA) model in [2], Sect. 6.3.2. It is assumed that two sensors obtain position measurements of the target with a sampling interval of $T = 1$ s. The standard deviation of the measurement noise is $\sigma_w = \sqrt{R_l} = 30$ m for each sensor (i.e., at each local tracker) and the process noise variance is $q = 1 \text{ m}^2/\text{s}^4$. T2TFwoM takes place every 5 s, i.e., at a reduced rate.

Table 1. Fuser variances (at fusion times) in steady state (fusion interval: 5 s)

Fusion Type	FC track at fusion time	
	Position	Velocity
T2TFwoMff	133	6.29
T2TFwoMpf	131	6.30
T2TFwoMnf	125	6.30
CTF	119	6.03
Single sensor tracker	205	7.26

Table 1 shows the steady state variances of position and velocity at the FC. All the fused tracks are more accurate than the single-sensor (local) tracks without fusion, which have steady state variances as 205 in position and 7.26 in velocity. Note that at the fusion time the position estimates of all the fused tracks have a small degradation compared to the CTF: 5% for T2TFwoMnf, 10% for T2TFwoMpf, 12% for T2TFwoMff. This shows that T2TFwoM has a degradation in fusion accuracy compared to CTF and this degradation increases in the presence of information feedback.

To explain this phenomenon consider the gains of the steady state filter for the above problem, namely, the alpha-beta filter. In steady state the filter gain is a *monotonically increasing function* of the maneuvering index λ [2].

Under T2TFwoMnf, the filter gain of each measurement in the fused track (with two equal-accuracy sensors) is

$$W^{\text{T2TFwoMnf}} = \frac{1}{2}[\alpha(\lambda_l), \beta(\lambda_l)/T]' \quad (1)$$

where $\lambda_l = \frac{\sqrt{q}T^2}{\sqrt{R_l}}$ is the local maneuvering index of the two trackers ($\sqrt{q}$ and $\sqrt{R_l}$ are the standard deviations of the process noise and measurement noise, respectively).

Under CTF, $z_c = \frac{1}{2}(z_1 + z_2)$ and $R_c = \frac{1}{2}R_l$. Thus, the central maneuvering index is $\lambda_c = \sqrt{2}\lambda_l$, i.e., larger. For each measurement, the centralized (globally optimal) filter gain in steady state for each measurement — its weighting — is

$$\begin{aligned} W^C &= \frac{1}{2}[\alpha(\lambda_c), \beta(\lambda_c)/T]' = \frac{1}{2}[\alpha(\sqrt{2}\lambda_l), \beta(\sqrt{2}\lambda_l)/T]' \\ &> W^{\text{T2TFwoMnf}} = \frac{1}{2}[\alpha(\lambda_l), \beta(\lambda_l)/T]' \end{aligned} \quad (2)$$

With information feedback, the local filter gains will be even smaller than without feedback, i.e., *they will deviate further* from the globally optimal gains.² This is because the local trackers have more accurate information due to the feedback (compared to the no feedback case) and this *reduces their filter gains* for the new measurements.

2.2 T2TF with Memory

In the configuration of T2TFwM, the fusion involves both the track estimates at the current fusion time and those from the previous fusion time. Fig. 6 illustrates T2TFwM with no, partial and full information feedback.

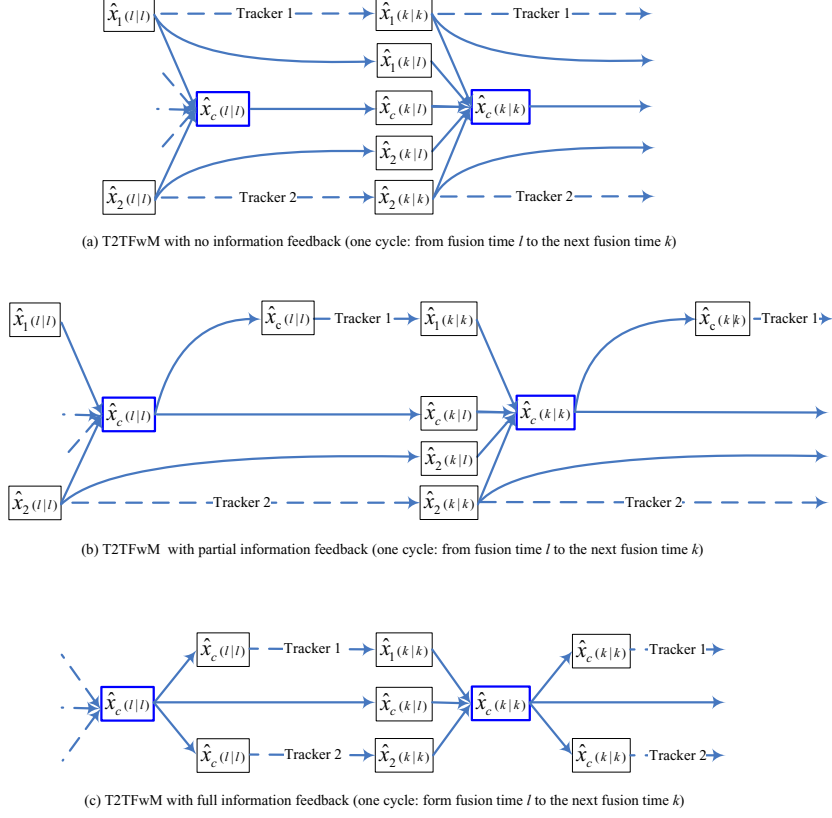
The exact fusion algorithms for the three T2TFwM configurations were presented in [16]. It was shown that, at full rate, T2TFwM has equivalent fusion performance with the CTF (for a linear system). However, at a reduced rate, compared to the CTF there is a certain loss of fusion accuracy, which is unavoidable [16]. And unlike the case with T2TFwoM, information feedback improves the fusion accuracy of T2TFwM. This phenomenon is illustrated with the following example.

The state of the target (taken as a scalar for simplicity) evolves according to

$$x(k) = x(k-1) + v(k) \quad k = 2, 3, \dots \quad (3)$$

where $v(k)$ is the process noise with variance $q = 0.3$.

² Gains smaller or larger than the optimal gains (which yield the minimum MSE) will lead to a MSE larger than the minimum [2]. The relationship between the optimal gain and the optimal state estimation MSE is discussed in detail in [2].

**Fig. 6.** T2TFwM at arbitrary rate

Two trackers, 1 and 2, take measurements of the target with independent measurement noises w_1 and w_2 , namely,

$$z_i(k) = x(k) + w_i(k) \quad i = 1, 2 \quad (4)$$

where $w_i(k)$ are zero-mean Gaussian noises with variance $R_i = 1, i = 1, 2$. The two trackers calculate tracks of the target with their own measurements using a Kalman filter. Each local track is initialized at time 1 with the first local measurement. The first T2TF happens at time 1. Then T2TFwM occurs every $N_f = 3$ sampling times.

Table 2 shows the fuser- and tracker-calculated variances when the fuser (with memory) is operating at reduced rate. These results verify the conclusions on the impact of information feedback on T2TFwM, namely, that feedback is beneficial. Note that due to the reduced rate, T2TFwM is suboptimal compared to CTF, but only slightly.

Table 2. Fuser and tracker 1 calculated variances at fusion times for $N_f = 3$ (reduced rate), $q = 0.3$, $R_1 = R_2 = 1$

Time		1	3	6	9	12	15
T2TFwMnf	Tracker1	1.0000	0.4639	0.4196	0.4180	0.4179	0.4179
	Fuser	0.5000	0.2772	0.2698	0.2694	0.2694	0.2694
T2TFwMpf	Fuser	0.5000	0.2763	0.2690	0.2688	0.2688	0.2688
T2TFwMff	Fuser	0.5000	0.2755	0.2683	0.2682	0.2682	0.2682
	CTF/CMF	0.5000	0.2743	0.2654	0.2653	0.2653	0.2653

Note that the IMF, detailed in the Sect. 4, also uses the previous track estimates (i.e., it has memory). When operating at full rate, IMF is algebraically equivalent to CTF and also to the algorithms for T2TFwM (for a linear system). However, at a lower rate, the IMF is heuristic.

Limitations of the exact T2TF algorithms discussed above include i) the exact fusion algorithms only exist under the Linear Gaussian assumption, ii) the algorithms require the crosscovariances of the tracks to be fused, which are generally difficult to obtain and greatly increase the complexity of the algorithms' implementation. For T2TF in practical tracking systems, approximate algorithms with near optimal fusion performance and less complexity are desirable. The IMF has been shown as a good candidate for the purpose and will be discussed in the next section.

3 Impact of Crosscorrelation on the Track-to-Track Fusion without Memory and No Feedback

A simple approximate approach for T2TF is to ignore the crosscovariances among different tracks. However this will lead to over-optimistic fused covariances, which, especially when full information feedback is used, could lead to large errors and filter divergence [16]. Here we use a simple example to show the effect of the crosscovariance on T2TF.

Two sensors tracking the same target, each with its local processor, are considered. The target is modelled by the kinematic model

$$x(k+1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x(k) + \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} v(k) \quad (5)$$

This is a nearly constant velocity (NCV) or white noise acceleration (WNA) motion model with unity sampling time with "piecewise constant" process noise (see [2], Sect. 6.3) with variance q entering into the system. A range of values for q will be considered.

The measurements at the two sensors are

$$z^m(k) = [1 \ 0] x(k) + w^m(k) \quad m = 1, 2 \quad (6)$$

with the two measurement noises mutually independent and with unity variance.

The maneuvering index for this target (see Eq. (6.5.3-14) in [2]), is thus

$$\lambda = \sqrt{q} \quad (7)$$

Fig. 7 presents in the first column the “ 2σ ” steady-state estimation error ellipses (corresponding to 85% probability mass under Gaussian assumption) in the state space x_2 - x_1 (velocity-position) *for each sensor*. The values of q considered are 0.01, 0.1, 0.5, 1, 2 and 5. These correspond to maneuvering indices from 0.1 to $\sqrt{5}$.

The second column presents the error ellipses one obtains by combining these estimates *assuming they are independent*. These ellipses are exactly half (in area) compared to the single-sensor error ellipses — they are, however, optimistic because the half variance (in each coordinate) is *unachievable*.

The third column presents the exact ellipses corresponding to the combined estimates obtained by *taking into account the dependence* between the two tracks due to the common process noise, quantified by the crosscovariance [6].

Fig. 8 shows the ratio of the areas of the ellipses of uncertainty of

- the fused estimates from the two sensors *accounting for their dependence* and
- the single sensor estimates

for various values of the target maneuvering index. This ratio would be equal to $1/2$ *in the absence of the dependence*. However, due to the dependence, it is around 0.7 for a wide range of process noise variances.

Fig. 9 shows the decreases in the elements of the covariance matrix for the same situation — as before, in the absence of dependence, each element would be halved after fusion.

For linear systems the crosscorrelations are positive and ignoring them leads to optimistic covariance for the fused estimates.

4 The Information Matrix Fusion

This section reviews the information fusion algorithm [10,8,15,11,6] and its extensions. The IMF operates similarly to the Information Matrix form of the KF — the Information Filter — and, consequently, it is designated as Information Matrix Fusion (IMF). The following versions of the IMF will be discussed, which are the IMF with full communication rate (IMF_{fc}), the IMF with reduced communication rate (IMF_{rc}) and the Generalized IMF (GIMF) for asynchronous T2TF (AT2TF) in the presence of communication delay.

4.1 IMF with Full Communication Rate – IMF_{fc}

The IMF, when operating at full rate, is equivalent to the optimal CTF [6]. The fused (central) estimate follows (for simplicity, N synchronized local trackers are assumed here)

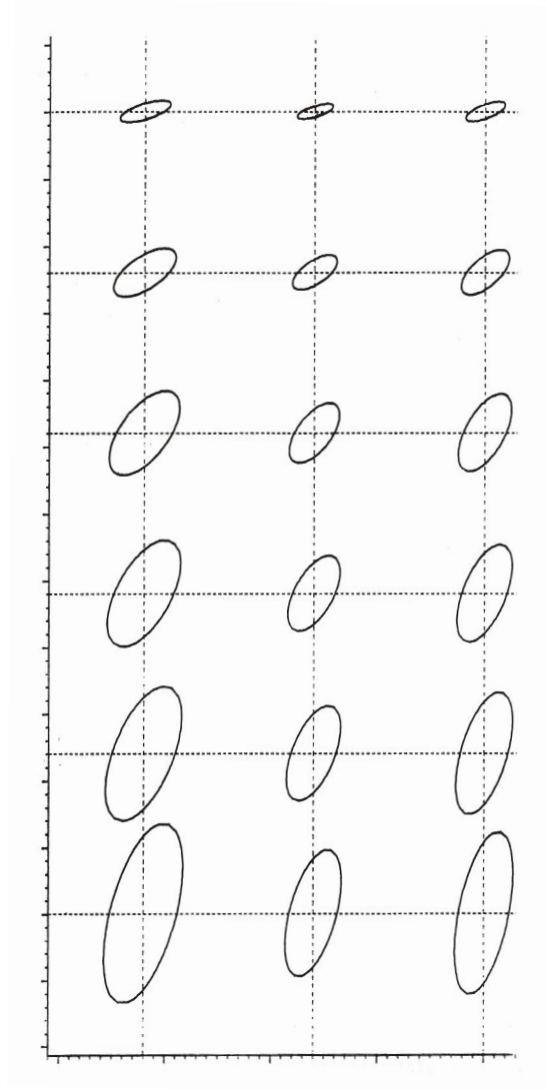


Fig. 7. Error ellipses for various levels of maneuvering index (Column 1: for each sensor alone; Column 2: for two sensors combined assuming independence; Column 3: for two sensors combined correctly)

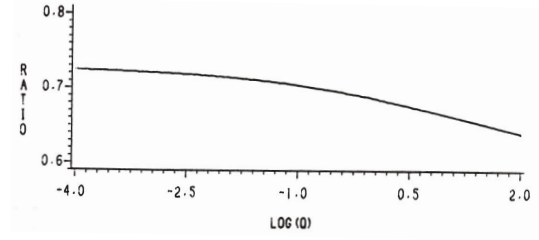


Fig. 8. Decrease in the error ellipse area in view of the dependence of the errors in the fusion process

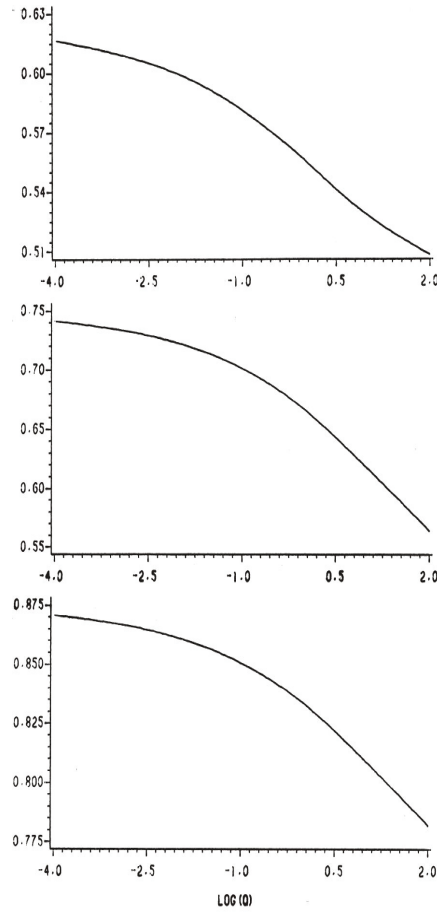


Fig. 9. Decrease of the state covariance matrix elements after fusion in view of the dependence of the errors

$$\begin{aligned}
P(k|k)^{-1}\hat{x}(k|k) &= P(k|k-1)^{-1}\hat{x}(k|k-1) \\
&+ \sum_{i=1}^N [P^i(k|k)^{-1}\hat{x}^i(k|k) - P^i(k|k-1)^{-1}\hat{x}^i(k|k-1)]
\end{aligned} \tag{8}$$

The updated fused covariance needed above is obtained as

$$P(k|k)^{-1} = P(k|k-1)^{-1} + \sum_{i=1}^N [P^i(k|k)^{-1} - P^i(k|k-1)^{-1}] \tag{9}$$

In IMFcr each local estimate/covariance has to be available at the center, i.e., *full communication rate* is necessary. The implementation of information feedback to local trackers is trivial in this configuration. Also note that the IMF is algebraically equivalent to the equivalent measurement approach [6].

4.2 IMF with Reduced Communication Rate – IMFcr

If the communication occurs only every n sampling times, equations (8)–(9) are used with the following modification. The fused (central) estimate follows from

$$\begin{aligned}
P(k|k)^{-1}\hat{x}(k|k) &= P(k|k-n)^{-1}\hat{x}(k|k-n) \\
&+ \sum_{i=1}^N [P^i(k|k)^{-1}\hat{x}^i(k|k) - P^i(k|k-n)^{-1}\hat{x}^i(k|k-n)]
\end{aligned} \tag{10}$$

The updated central covariance needed above is obtained in terms of the local covariances as

$$P(k|k)^{-1} = P(k|k-n)^{-1} + \sum_{i=1}^N [P^i(k|k)^{-1} - P^i(k|k-n)^{-1}] \tag{11}$$

It should be noted that the above is *no longer equivalent to the CTF* — the modified equations (10)–(11) are heuristic. As shown in [9], with full information feedback, the IMFcr diverges for extremely large values of process noise variance. However, for practical levels of process noises, the IMFcr was shown to have consistent and close optimal fusion performance. Compared to the exact T2TF fusion algorithms, the IMFcr does not require the evaluation of the crosscovariances between the tracks to be fused, which significantly simplifies the implementation and makes it a good candidate for T2TF in practical tracking systems. Next the generalization of the IMF for the problem of asynchronous T2TF will be discussed.

4.3 Generalized Information Matrix Fusion for Asynchronous T2TF

The T2TF algorithms mentioned above assume that the local tracks are synchronized. In practical distributed tracking systems, the synchronicity assumption

can hardly be satisfied, which raises the problem of asynchronous track-to-track fusion (AT2TF). In addition local tracks arrive at the FC with transmission delays, which further complicates the fusion problem and the implementation of information feedback. To address this problem, a generalized IMF (GIMF) for AT2TF was presented in [18], and is reviewed next.

Without loss of generality consider the fusion of tracks from two asynchronous local trackers 1 and 2. Tracker 1 is collocated with the FC (no communication delay and information feedback to tracker 1). Tracker 2 is a remote tracker (with communication delay; no feedback to it). Suppose at the fusion time t_f one has

- track $(\hat{x}^1(t_f|t_f), P^1(t_f|t_f))$ from tracker 1 (same as FC) and
- tracks $(\hat{x}^2(t_1|t_1), P^2(t_1|t_1))$ and $(\hat{x}^2(t_2|t_2), P^2(t_2|t_2))$ from tracker 2, $t_1 < t_2 \leq t_f$,

where t_1 and t_2 are the previous and current communication times from sensor 2.

According to the Generalized Information Matrix fusion (GIMF) the fused track is given by

$$P(t_f)^{-1} = P^1(t_f|t_f)^{-1} + [P^2(t_f|t_2)^{-1} - P^2(t_f|t_1)^{-1}] \quad (12)$$

$$P(t_f)^{-1}\hat{x}(t_f) = P^1(t_f|t_f)^{-1}\hat{x}^1(t_f|t_f) + [P^2(t_f|t_2)^{-1}\hat{x}^2(t_f|t_2) - P^2(t_f|t_1)^{-1}\hat{x}^2(t_f|t_1)] \quad (13)$$

where $\hat{x}(t_f)$ is the fused track at t_f , $P(t_f)$ is its covariance, $\hat{x}^1(t_f|t_f)$ and $P^1(t_f|t_f)$ are the track and its covariance from tracker 1 at the fusion time t_f , $\hat{x}^2(t_f|t_i)$ and $P^2(t_f|t_i)$ are the predicted local track 2 from t_i to the fusion time t_f and the corresponding covariance, $i = 1, 2$.

In the presence of communication delay, information feedback to the remote tracker 2 needs to be carefully handled. See [18] for the details of the implementation for both AT2TF with partial and AT2TF with full information feedback. It was shown that the proposed GIMF based AT2TF algorithms yield consistent and close to optimal fusion results. The following reasons contribute to the applicability of the GIMF:

- The predicted information gain from track 2 quantified by $[P^2(t_f|t_2)^{-1} - P^2(t_f|t_1)^{-1}]$ in (12), is due to the local measurements from $(t_1, t_2]$ and can be viewed as approximately independent from the other tracks.
- The subtraction structure of the information gain $[P^2(t_f|t_2)^{-1} - P^2(t_f|t_1)^{-1}]$ provides a desirable feature that cancels (approximately) its crosscorrelation with other tracks caused by the common process noises with the use of prediction.

5 Heterogenous Track-to-Track Fusion

The previously discussed T2TF configurations and algorithms assume that the local trackers use the same target state vector. In practical tracking systems, local

trackers may use different motion models and state vectors, due to observability issues (e.g., active vs. passive sensors).

A concrete example of such a situation is when (i) tracker 1 uses an active sensor which is able to obtain range and azimuth measurements (full 2D or 3D position, the latter requires elevation measurements as well) and its target state vector comprises Cartesian position, velocity, etc. (ii) tracker 2 uses a passive sensor with angle only measurements and its target state vector comprises angular position, velocity and possibly acceleration.

5.1 The Heterogenous Track-to-Track Fusion Problem

Without loss of generality, consider the following state-space models

$$\mathbf{x}^i(k+1) = f^i[\mathbf{x}^i(k)] + \mathbf{v}^i(k) \quad (14)$$

$$\mathbf{z}^i(k) = h^i[\mathbf{x}^i(k)] + \mathbf{w}^i(k) \quad (15)$$

at sensor i and

$$\mathbf{x}^j(k+1) = f^j[\mathbf{x}^j(k)] + \mathbf{v}^j(k) \quad (16)$$

$$\mathbf{z}^j(k) = h^j[\mathbf{x}^j(k)] + \mathbf{w}^j(k) \quad (17)$$

at sensor j . In the above, $f^s[\cdot]$ and $h^s[\cdot]$, $s = i, j$, are different and can be nonlinear; $\mathbf{v}^s(\cdot)$ and $\mathbf{w}^s(\cdot)$, $s = i, j$, are the process and measurement noises, respectively.

The state vectors $\mathbf{x}^i$ and $\mathbf{x}^j$ are in different spaces. Let $\mathbf{x}^i$ be the larger dimension state. For example, one can have full Cartesian position and velocity in 2-dimensional space for tracking with an active sensor

$$\mathbf{x}^i = [x \ \dot{x} \ y \ \dot{y}]' \quad (18)$$

and $\mathbf{x}^j$ be the smaller dimension state (e.g., angular position and velocity for tracking with a passive sensor)

$$\mathbf{x}^j = [\theta \ \dot{\theta}]' \quad (19)$$

These state vectors, in general, have the nonlinear relationship

$$\mathbf{x}^j \triangleq g(\mathbf{x}^i) \quad (20)$$

The two sensors are assumed synchronized³ and the time index k for sampling time t_k will be omitted if there is no ambiguity.

³ Generalization to asynchronous sensors is possible [17], but the notations become very cumbersome. Without considering the crosscovariance matrix, the extension to asynchronous case is straightforward. If the crosscovariance matrix is considered (for the configuration with no memory at the FC and no information feedback to the local trackers), each track's latest estimate available at the FC is predicted to the fusion time and then they are fused using the appropriate covariance matrices.

The corresponding estimates (approximate conditional means) at these heterogeneous local sensors are $\hat{\mathbf{x}}^i$ with (conditional) covariance matrix (the conditioning is omitted for brevity)

$$P^i \triangleq E[(\mathbf{x}^i - \hat{\mathbf{x}}^i)(\mathbf{x}^i - \hat{\mathbf{x}}^i)'] \quad (21)$$

and $\hat{\mathbf{x}}^j$ with (conditional) covariance matrix

$$P^j \triangleq E[(\mathbf{x}^j - \hat{\mathbf{x}}^j)(\mathbf{x}^j - \hat{\mathbf{x}}^j)'] \quad (22)$$

The problem is how to carry out the fusion of the estimate $\hat{\mathbf{x}}^i$ with covariance P^i and the estimate $\hat{\mathbf{x}}^j$ with covariance P^j to achieve better estimation performance for the full state of interest $\mathbf{x}^i$.

5.2 The LMMSE Fuser

The approach to HT2TF discussed next is to use the linear technique based on the fundamental equations of LMMSE estimation [2]. Considering the full state estimate $\hat{\mathbf{x}}^i$ as the prior and the smaller state estimate $\hat{\mathbf{x}}^j$ as the measurement, we have the LMMSE fused estimate

$$\hat{\mathbf{x}}_{\text{LMMSE}}^i = \hat{\mathbf{x}}^i + P_{\mathbf{xz}} P_{\mathbf{zz}}^{-1} [\hat{\mathbf{x}}^j - g(\hat{\mathbf{x}}^i)] \quad (23)$$

with the corresponding fused covariance matrix

$$P_{\text{LMMSE}}^i = P^i - P_{\mathbf{xz}} P_{\mathbf{zz}}^{-1} P_{\mathbf{xz}}' \quad (24)$$

where (the details can be found in [19])

$$\begin{aligned} P_{\mathbf{xz}} &\triangleq E[(\mathbf{x}^i - \hat{\mathbf{x}}^i)(\hat{\mathbf{x}}^j - g(\hat{\mathbf{x}}^i))'] \\ &\approx P^i (G^i)' - P^{ij} \end{aligned} \quad (25)$$

$$\begin{aligned} P_{\mathbf{zz}} &\triangleq E[(\hat{\mathbf{x}}^j - g(\hat{\mathbf{x}}^i))(\hat{\mathbf{x}}^j - g(\hat{\mathbf{x}}^i))'] \\ &\approx P^j - G^i P^{ij} - P^{ji} (G^i)' + G^i P^i (G^i)' \end{aligned} \quad (26)$$

with G^i the Jacobian of $g(\mathbf{x}^i)$

$$G^i \triangleq [\nabla_{\mathbf{x}^i} g(\mathbf{x}^i)']_{\mathbf{x}^i = \hat{\mathbf{x}}^i} \quad (27)$$

and P^{ij} the crosscovariance matrix

$$P^{ij} \triangleq E[(\mathbf{x}^i - \hat{\mathbf{x}}^i)(\mathbf{x}^j - \hat{\mathbf{x}}^j)'] \quad (28)$$

5.3 The ML Fuser

Under the Gaussian assumption, the heterogeneous T2TF problem can be solved by minimizing the negative log-likelihood function⁴ (NLLF) based on (20). The LF of $\mathbf{x}^i$ (the full state) is [6]

$$\Lambda(\mathbf{x}^i) = p(\hat{\mathbf{x}}^i, \hat{\mathbf{x}}^j | \mathbf{x}^i) = \mathcal{N} \left\{ \begin{bmatrix} \hat{\mathbf{x}}^i \\ \hat{\mathbf{x}}^j \end{bmatrix}; \begin{bmatrix} \mathbf{x}^i \\ \mathbf{x}^j \end{bmatrix}, P^{-1} \right\} \quad (29)$$

where

$$P = \begin{bmatrix} P^i & P^{ij} \\ P^{ji} & P^j \end{bmatrix} \quad (30)$$

Then the ML fused estimate is the solution of

$$\nabla_{\mathbf{x}^i} L(\mathbf{x}^i) = 0 \quad (31)$$

where

$$L(\mathbf{x}^i) \triangleq -\ln \Lambda(\mathbf{x}^i) \quad (32)$$

Because of the nonlinearity of the function $g(\mathbf{x}^i)$, there is no explicit expression for the solution of (31). Consequently, (32) is minimized by a numerical search, e.g., the gradient projection algorithm. The result is denoted as $\hat{\mathbf{x}}_{\text{ML}}^i$ and the corresponding covariance matrix is

$$P_{\text{ML}}^i = \left(\begin{bmatrix} I & G^i \end{bmatrix} P^{-1} \begin{bmatrix} I \\ G^i \end{bmatrix} \right)^{-1} \quad (33)$$

where G^i is defined in (27) and I is the identity matrix (4×4 in our case).

6 Example

Fig.10 shows an example scenario (see [19] for details), where the active sensor is located at $(-6 \times 10^4, 2 \times 10^4)$ m with sampling interval $T_a = 5$ s and the passive sensor located at $(-5 \times 10^4, 4 \times 10^4)$ m with sampling interval $T_p = 1$ s. Measurement noises from the two sensors are assumed to be mutually independent zero mean white Gaussian noises with standard deviations $\sigma_r = 20$ m, $\sigma_a = 5$ mrad for the active sensor, and $\sigma_p = 0.5$ mrad for the passive sensor.

The tracker at the active sensor uses an IMM estimator with two modes: mode 1, a linear nearly constant acceleration (NCV) model [2], and mode 2, a nonlinear nearly coordinate turn (NCT) model [2]. The tracker at the passive sensor uses a linear KF (rather than IMM estimator, because target maneuvers are practically unobservable by the passive sensor). The motion model is the discretized continuous Wiener process acceleration (CWPA) model (with angle, angle rate and angle acceleration).

In [19] it was observed that for this (nonlinear) HT2TF problem

⁴ As it is pointed out in [8], the LMMSE T2TF approach is, in the linear Gaussian case, optimal in ML sense.

- The crosscorrelation of the estimation errors from heterogeneous local trackers is too complicated to capture — it can be *positive or negative*. The estimation errors' crosscorrelation has been examined by MC simulations.
- When using a LMMSE fuser, neglecting the track crosscovariance in HT2TF leads to sometimes optimistic, sometimes pessimistic fused covariance. Note this is different from the homogenous T2TF where neglecting the crosscovariance between the local tracks will always result in optimistic fused covariance. This is because for linear systems the crosscorrelation coefficients are always positive.
- When the configuration of fusion without memory and no information feedback is used, neglecting the track crosscovariance is a reasonable practical choice, which yields little loss in fusion performance.

The results using $\hat{\mathbf{x}}_{\text{LMMSE}}^i$ with P_{LMMSE}^i and $\hat{\mathbf{x}}_{\text{ML}}^i$ with P_{ML}^i were found to be practically the same so only the former is compared with the CTF which processes all the measurements (from both the active and the passive sensor) in the FC.

Fig. 11 compares the position RMSE of the HT2TF algorithm to that of the CTF IMM tracker from 400 Monte Carlo runs. Surprisingly, the results show that, in the scenario considered, the HT2TF is superior to the centralized IMM tracker (CTF IMM) during the maneuver periods when the latter experiences a spike in its error.

To explain this apparently counterintuitive result, Fig. 12 shows the maneuvering mode probabilities (NCT) in the active sensor IMM and CTF IMM. It turns out that the use of the passive measurements in the CTF IMM “clouds” the maneuvers because of low maneuvering index, which leads to the degraded

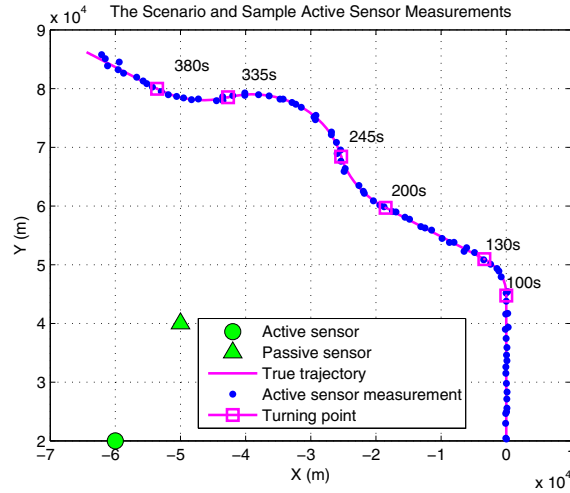


Fig. 10. The scenario, with the target true speed 250 m/s

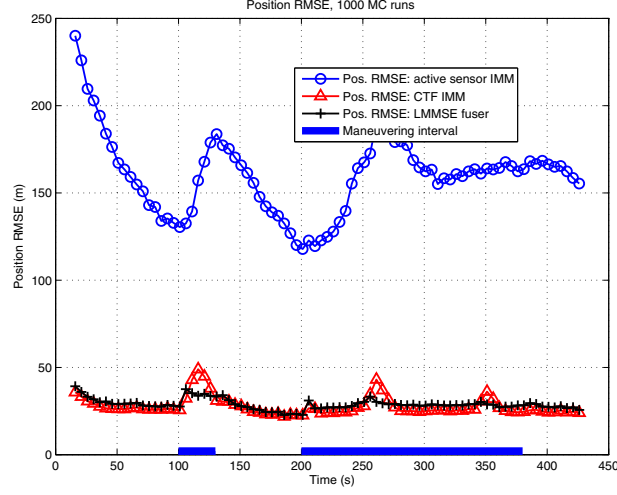


Fig. 11. Performance comparison: HT2TF (with LMMSE fuser) vs. CTF IMM in position RMSE

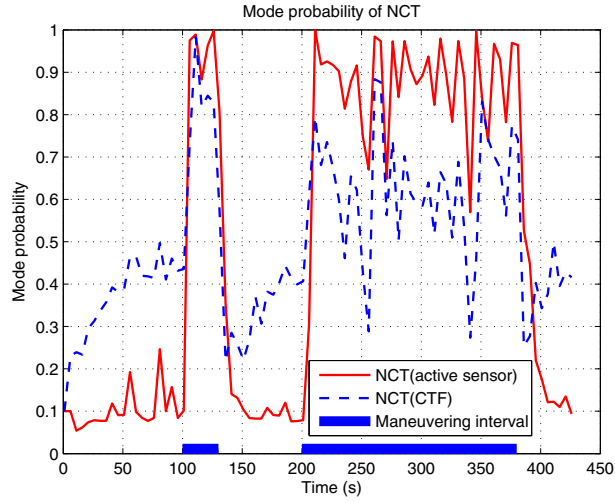


Fig. 12. Maneuvering mode probability (NCT) in the active sensor's IMM and CTF IMM

performance of the CTF IMM filter. Specifically, as can be seen from Fig. 12, when there is a maneuver, the maneuvering mode probability in the CTF IMM rises slower and to a lower level than in the active sensor's IMM. (Note that in this example the passive sensor has a higher sampling rate than the active sensor with $T_a = 5$ s and $T_p = 1$ s.) In the scenario considered, the freedom available to

each local sensor to flexibly design a more suitable local estimator is what allows the heterogeneous T2TF approach to achieve a better estimation performance than the CTF IMM.

7 Conclusions

This chapter discusses the various architectures for track-to-track fusion (T2TF). Based on whether the track estimates from the previous fusion are used by the current fusion and the presence of information feedback, T2TF can be further categorized into six configurations, namely, T2TF without memory with no, partial and full information feedback, and T2TF with memory with no, partial and full information feedback.

T2TF without Memory (T2TFwoM) uses only the latest local tracks. It has a small loss in fusion accuracy compared to the centralized tracker/fuser (CTF) regardless of fusion rate. It was shown that information feedback has a negative impact on the accuracy of T2TFwoM.

T2TF with Memory (T2TFwM) uses both the latest local tracks and the track estimates from the previous fusion. It is algebraically equivalent to the CTF when operating at full rate. At reduced rates, it has a slight loss in fusion accuracy compared to the CTF. In contrast to T2TFwoM, information feedback has a positive impact on the fusion accuracy of T2TFwM.

The Information Matrix Fusion (IMF) is a special form of T2TFwM. It is equivalent to the CTF at full rate. At reduced rates it is heuristic but performs well for practical levels of process noises: it has near optimal fusion performance and is consistent. For the fusion of asynchronous tracks in the presence of communication delay, fusion algorithms were developed based on a generalized IMF, which are easy to implement, support the use of information feedback and yield consistent, close to optimal fusion results.

The Heterogenous Track-to-Track Fusion (HT2TF) problem involves the fusion of tracks in different state spaces. Unlike in the conventional T2TF, the crosscorrelations of local tracks have indefinite impact on the fuser-calculated accuracy of the fused track. When an IMM tracker is used, HT2TF allows each local sensor to flexibly design a more suitable local estimator which can lead to a better estimation performance than the CTF.

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